In [1]:
```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

Initial Data Analysis

In [2]:
```
df = pd.read_csv("playgolf_data.csv")
df
```

Out[2]:

	Outlook	Temperature	Humidity	Wind	PlayGolf
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rainy	Mild	High	Weak	Yes
4	Rainy	Cool	Normal	Weak	Yes
5	Rainy	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rainy	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rainy	Mild	High	Strong	No

In [3]:
```
df.shape
```

Out[3]:
```
(14, 5)
```

In [4]:
```
df.dtypes
```

Out[4]:
```
Outlook        str
Temperature    str
Humidity       str
Wind           str
PlayGolf       str
dtype: object
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Outlook        0
Temperature    0
Humidity       0
Wind           0
PlayGolf       0
dtype: int64
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

In [7]:
```
df.describe(include='all')
```

Out[7]:

	Outlook	Temperature	Humidity	Wind	PlayGolf
count	14	14	14	14	14
unique	3	3	2	2	2
top	Sunny	Mild	High	Weak	Yes
freq	5	6	7	8	9

In [8]:
```
num_cols = df.select_dtypes(include="number").columns.tolist()
cat_cols = df.select_dtypes(exclude="number").columns.tolist()
target = "PlayGolf"
print("Numerical:", num_cols)
print("Categorical:", cat_cols)
print("Target:", target)
```

```
Numerical: []
Categorical: ['Outlook', 'Temperature', 'Humidity', 'Wind', 'PlayGolf']
Target: PlayGolf
```

In [9]:
```
df[target].value_counts()
```

Out[9]:
```
PlayGolf
Yes    9
No     5
Name: count, dtype: int64
```

Data Pre-processing

In [10]:
```
df = df.dropna().drop_duplicates()
df.shape
```

Out[10]:
```
(14, 5)
```

In [11]:
```
features = [c for c in df.columns if c != target]
features
```

Out[11]:
```
['Outlook', 'Temperature', 'Humidity', 'Wind']
```

In [12]:
```
X = pd.get_dummies(df[features], drop_first=True).astype(int)
y = df[target]
X.head()
```

Out[12]:

	Outlook_Rainy	Outlook_Sunny	Temperature_Hot	Temperature_Mild	Humidity_Normal	Wind_Weak
0	0	1	1	0	0	1
1	0	1	1	0	0	0
2	0	0	1	0	0	1
3	1	0	0	1	0	1
4	1	0	0	0	1	1

Train-Test Split

In [13]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print(X_train.shape, X_test.shape)
```

```
(11, 6) (3, 6)
```

Decision Tree Model

In [14]:
```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

Out[14]:

▾ DecisionTreeClassifier

▸ Parameters

▸ Fitted attributes

In [15]:
```
y_pred = model.predict(X_test)
y_pred
```

Out[15]:
```
array(['Yes', 'No', 'No'], dtype=object)
```

Evaluation

In [16]:
```
accuracy_score(y_test, y_pred)
```

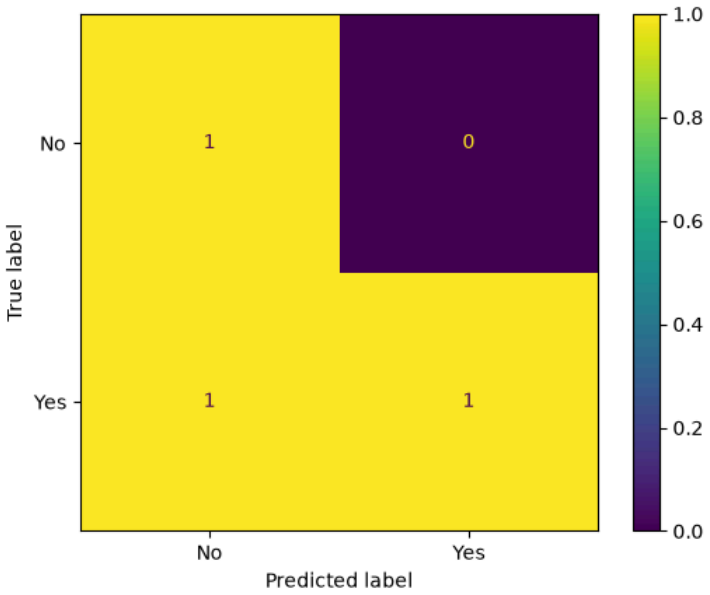
Out[16]:
```
0.6666666666666666
```

In [17]:
```
cm = confusion_matrix(y_test, y_pred)
pd.DataFrame(cm, index=["Actual No", "Actual Yes"], columns=["Predicted No", "Predicted Yes"])
```

Out[17]:

	Predicted No	Predicted Yes
Actual No	1	0
Actual Yes	1	1

In [18]:
```
ConfusionMatrixDisplay(cm, display_labels=['No', 'Yes']).plot()
plt.show()
```



In [19]:
```
comparison = pd.DataFrame({"Actual": y_test.values, "Predicted": y_pred})
comparison
```

Out[19]:

	Actual	Predicted
0	Yes	Yes
1	Yes	No
2	No	No

In [20]:
```
plt.figure(figsize=(14, 8))
plot_tree(model, feature_names=X.columns, class_names=model.classes_, filled=True)
plt.show()
```

